

SQL_Project2(아마존_판매_데이터셋_최홍주)

Amazon 데이터셋 기반 추천 시스템 설계 프로젝트

프로젝트 개요

이번 프로젝트는 **Amazon** 데이터셋을 기반으로 **고객 맞춤형 추천 시스템**을 설계하고 구현하는 작업입니다. Amazon은 전 세계적으로 가장 큰 전자상거래 플랫폼 중 하나로, 방대한 고객 리뷰와 제품 데이터를 보유하고 있습니다.

이번 프로젝트에서는 이러한 데이터를 SQL만을 사용하여 분석하고, **고객의 구매 경험을 향상시킬 수 있는 추천 시스템**의 기초를 설계하는 것이 목표입니다.

학습 목표

데이터 이해 및 탐색

먼저 Amazon 데이터셋을 분석하여 **고객과 제품 간의 상호작용 데이터**를 이해합니다. 고객 리뷰, 평점, 제품 카테고리 등 다양한 데이터를 활용해 주요 인사이트를 도출합니다.

추천 시스템 설계 및 구현

SQL을 사용해 인기 제품 추천, 고객 선호도를 기반으로 한 추천, 유사 제품 추천 등 다양한 추천 셋을 제안할 겁니다. 여러분은 동일한 데이터를 다양한 각도에서 분석해보면서 새로운 서비스에 활용할 수 있는 가능성을 분석적으로 보여주시면 됩니다.

주요 개념

필드명	설명
product_id	제품의 고유 식별자
product_name	제품의 이름
category	제품이 속한 카테고리 정보
discounted_price	할인된 가격
actual_price	정가
discount_percentage	할인율
rating	제품의 평균 평점
rating_count	제품에 대한 총 평점 수
about_product	제품에 대한 간단한 설명
user_id	리뷰를 작성한 사용자의 고유 식별자 (선택으로 구분된 여러 사용자)
user_name	리뷰를 작성한 사용자 이름 (선택으로 구분된 여러 사용자)
review_id	리뷰의 고유 식별자 (선택으로 구분된 여러 리뷰)
review_title	리뷰 제목 (선택으로 구분된 여러 리뷰 제목)
review_content	리뷰 내용 (선택으로 구분된 여러 리뷰 내용)
img_link	제품 이미지 URL
product_link	제품 페이지 URL

- 전체 데이터 몇 행 몇 컬럼? **1465행 / 16컬럼**
- 데이터 타입

☐	필드 이름	유형	모드
☐	product_id	STRING	NULLABLE
☐	product_name	STRING	NULLABLE
☐	category	STRING	NULLABLE
☐	discounted_price	FLOAT	NULLABLE
☐	actual_price	FLOAT	NULLABLE
☐	discount_percentage	FLOAT	NULLABLE
☐	rating	STRING	NULLABLE
☐	rating_count	INTEGER	NULLABLE
☐	about_product	STRING	NULLABLE
☐	user_id	STRING	NULLABLE
☐	user_name	STRING	NULLABLE
☐	review_id	STRING	NULLABLE
☐	review_title	STRING	NULLABLE
☐	review_content	STRING	NULLABLE
☐	img_link	STRING	NULLABLE
☐	product_link	STRING	NULLABLE

- 데이터 헤드

일	product_id	product_name	category	discounted_price	actual_price	discount_percent...	rating	rating_count
1	B0BFBKX9F	Khaffan Orfin Fan heater for Home and Kitchen-KD 2215	Home&KitchenHeatingCoolin gAirQualityRoomHeaterStfa rtheaters	1299.0	2495.0		0.48	2
2	B0BFBKX9F	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice, Shakes, and Smoothies (Pink)	Home&KitchenKitchen&Hom eAppliancesSmallKitchenAppl iancesHandBlenders	669.0	1499.0		0.55	23
3	B0B7L8VYCB	Green Teles Heat Seal Mini Food Sealer Impulse Machine for Sealing Plastic Bags Packaging	Home&KitchenKitchen&Hom eAppliancesSmallKitchenAppl iancesVacuumSealers	161.0	300.0		0.46	24
4	B0BNDGL2ST	MR. BRAND Portable USB Juicer Electric USB Juice Maker Bottle Blender Grinder Mixer 6 Blades Rechargeable Bottle with (Multi color) (MULTI MIXER 6 BLD)	Home&KitchenKitchen&Hom eAppliancesSmallKitchenAppl iancesJuicerMixerGrinders	499.0	2199.0		0.77	2.6

about_product	user_id	user_name	review_id	review_title	review_content	img_link	product_link
Home and KitchenPOWERFUL 2000 WATT HEATING POSITION 1000 W2000WADJUSTABLE ✓ [Easy to carry around] - This handheld blender is equipped with a travel cover for easy carrying. You can drink nutritious juices, Sealing Machine use for snack food cover storage and creates airtight containers, make food fresh, Heat Sealer will easily reseal original Portable & Convenient to Charge: This little handy light weight design is easy for you carry it to school, office, parks, camping, anywhere you want	ASHT3KAKG35MAQQUOR8T5V...	Mandipa Sengupta,Vishwa	R10C2DE615EX1JRH6SLTJM...	Bad quality,Amazing product...	The heating capacity is zero Moreover I have not send return request. Noone has come to collect it.Best heater at this price. Quality is very	https://m.media-amazon.com /images/W/WEBP_402278-71 /images/I/STB8P415mLL_SX 300_SV300_QL_75_J_Mwelo_p g	https://www.amazon.in/Khaff an-Orfin-Fan-Heater-Home-Kitch e-2215/dp/B0BFBKX9F/ref=pa _t_328/qid=167292367&as in=1-328
	AH9DA6AKDSXYCEZJANW4E UJN8PQJALZSSOOLBIMV UTUK3FTTQ5MAG5ANAVCL8 CPVQ3MYPTU8F32N7W5A, AE76J2OUQ9PW684QHTL	Mariya,Been,PooB Parasarani,R...	R1WJBT3U9PCULRM9H8FX 9450201846X7NLSM4VY1 827M20D8W68J11N6Q2R0 KARUJ	Now product I recommend to buy do not buy! Defective product.Wast of money,very very bad portable,Waste of money its not working	I liked that it is so convenient to carry, Wast of money Defective product, cheap quality, doesn't blend at all!https://m.media-amazon.c	https://m.media-amazon.com /images/I/4177Q2uulL_SX3 05_SV300_QL_75_J_Mwelo_p g	https://www.amazon.in/Perio nal-Blender-Portable-Sanitary-6 /dp/B0BFBKX9F/ref=pa _t_328/qid=167292367&as in=1-328
	AG2B89Q2AG87BF8UR0Y5 MFD6A5AGP03AS4VWYV5 ANWHTU7FETZ5AAZDPCSL 34FZ20M6K0H0L6KQKAF DS7H05SL34CZ8N07NG4X	Narvin Thawani,Amazon Customer,Jemba Lawrence,Aditya Kunal Kapoor,Mangamala Suresh,Juhi sharma,nagis	R5M9MH8DCTJBMJ9J2DDB8 LWFJ2JL6J0W50W20325A, R3J3JLJ0M6W9WYJSP8E3D 3D5LLW0J9182152ZSMW42 J11AK93TEZ5X1AR1T0DMH	It is broken.Could have been a bit better.The first one was not working and the replacement was sent without box with a used piece,Super	It is broken and it is not useful. Please change this product,product is so so... the plastic quality is not so good as per the price.it claims that	https://m.media-amazon.com /images/W/WEBP_402278-72 /images/I/411N81EXJNL_SV 300_SX300_QL_75_J_Mwelo_p g	https://www.amazon.in/Green -Teles-Sealer-Impulse-Machin e/dp/B0B7L8VYCB/ref=pa _t_383/qid=167292367&as in=1-383
	AEJXHA3AMVUZVWQZD84 B84GT4AP8BF803BULG63 6GZ2Z2CM7ALASD3C8R8B DVGM737YYPNCE07XQJAH RS3W5LAAG6FKD8WRMX	Yashwant Gots,Nargis Sulhasa,S.UMAABDeLalates h Hegde,Adhithy Aravind,Any babee,Sande penam	R8B7U4WZB8WQJ93V1H6 B8H6ALJ25N6307N6SBLR 38C31JLJN8X9SJBH4M5Q208 A06B82DQ9H14M69TVAJ2P QHLS0100F4R5TYV665P2	Portable but not much powerful,not so good, joverit back up is very poor, its more like a toy for my 9 years old girl, 600 rs ok,Don't be fooled	Not sufficient power,the product is more like a toy for kids, can't use it as a proper juicer, power back up is very poor,Ok,i had to replace this	https://m.media-amazon.com /images/I/37P585D4L_SV3 05_SX300_QL_75_J_Mwelo_p g	https://www.amazon.in/MR.B ROAD-Portable-Electric-Heate r/dp/B0BNDGL2ST/ref=pa _t_1_302/qid=167292367&as in=1-302

데이터 전처리

불필요 컬럼 제거(16컬럼 → 14컬럼)

- **img_link**, **product_link** : 두 컬럼은 분석에 불필요하다 판단되어 제거했습니다.

```
CREATE OR REPLACE TABLE `modulabs_project.amazon` AS
SELECT
  product_id,
  product_name,
  category,
  discounted_price,
  actual_price,
  discount_percentage,
  rating,
  rating_count,
  about_product,
  user_id,
  user_name,
  review_id,
  review_title,
```

```
review_content
FROM `modulabs_project.amazon_raw`
```

결측치 처리

행	COUNT_product_id	COUNT_product...	COUNT_category	COUNT_discount...	COUNT_actuaL_pr...	COUNT_discount...	COUNT_rating
1	1465	1465	1465	1465	1465	1465	1465

COUNT_rating_co...	COUNT_about_pr...	COUNT_user_id	COUNT_user_name	COUNT_review_id	COUNT_review_ti...	COUNT_review_c...
1463	1465	1465	1465	1465	1465	1465

- rating_count 결측치 확인 → 결측값 2행 존재했으나, 실제 리뷰 데이터가 1개씩 있음 → 1로 채우기로 결정

rating_count	about_product	user_id	user_name	review_id	review_title
null	USB C to C Cable: This cable has type C connector at both ends and a USB C type wall charger/adapter will be	AE7CFHY23VA...	Pranav	RUB7U91HVZ30	The cable works but is not 55W as advertised
null	[The Fastest Charge] - This iPhone USB C cable supports PD 3.0 fast charging, up to 20W with USB-C Power	AGJCS05H5BB...	Abdul Gafur	RQXD5SAMP06L	Awesome Product

```
UPDATE `modulabs_project.amazon`
SET rating_count = 1
WHERE rating_count IS NULL
```

행	COUNT_rating_count
1	1465

중복 데이터 확인

- 완전 동일 행 중복 여부: 전체 55행(완전 동일 행이 각각 2~3개 존재함)

```
SELECT *, COUNT(*) as cnt
FROM `modulabs_project.amazon`
GROUP BY ALL
HAVING cnt > 1
```

작업 정보	결과	시각화	JSON	실행 세부정보	실행 그래프
user_name	review_id	review_title	review_content	cnt	
Amazon	RWSHFGBE1WU3R1VBNT43	Its slow in charging,Ok	Charging power is very less. It takes hours to charge even a single phone.,Ok type product.	2	
Customer,DHONDU,Saurabh,romod George,Ramesh	HSMVMB,RTATA9H2ELJ61,R1B0APD6HVOT8VR99TNL1C7	product,Looks good, but charges slowly very slow charging.,Poor quality, iPhone part will last for two months	Not very sturdy,Although it is written fast charging.		
Harar,mram,Placeholder,SHASHANK	XQ50,R37RT17N8YUWT4,R1WG1ARVL9YH61,R2UFM5PK06	Will not work with new system,Veri good,Ok	Was working fine with window 10 old computer but is not installable with new system on Window 11,Overall very good item,Easy to install ok	2	
SQLARUN KUMAR,veer,Ravi shankar,Shivam	R3AZDEK3MQA3RA,RXF3HOC	product,Access wifi signal.,very good,Good Product,8139EU based	This connector has provided as a boon for my old Samsung LCD monitor and I am proactively using that monitor as the 2nd screen just if the	2	
Kumac,ramesh,Bibin,Hem Roy	BWVVOVB,R6CVYFDUXBS36,R1QMN1WQJWAB7,R2MOVGG	Good product ; Average Finishing,Save it purpose well without any issue. I am satisfied,No issues,NO NEED TO PUT AUX CABLE FOR	This was a saviour for me as I didn't want to buy expensive	2	
aditya d,Paranthaman,ashik rahman c,Rajas c.,Madan Tejaswi,Sunil,chailanya srinikrishna,subeh sharma	R1GYK05N67470,R1J21BZ29NGQF9,R16JCHEILBYOMWR	Only affordable Stylius that works with Apple,Product is			
2Z542 TWX	2WVVVS88M7SH18,R2MQ3VB8ZTUS48,RBJPTKHQY7G7U,R37PK05FURJW35,R38R2YC2J				
JKRE	R2FRXL54FATWQ,ROBDUJXECNYM,R6GD9MATBB0D,RG				
4EEIQ					
K50Z					

- 완전 동일 행 제거 후 테이블 업데이트(1465행 → 1400행)

```
CREATE OR REPLACE TABLE `modulabs_project.amazon` AS
SELECT DISTINCT *
FROM `modulabs_project.amazon`
```

행	product_id	product_name	category	discounted_price	actual_price	discount_percent...
1	B0BPJBT83F	Khaitan ORFIn Fan heater for H...	Home&KitchenHeating.Coolin g&AirQualityRoomHeaters(Fa nHeaters	1299.0	2495.0	0.48
2	B0BFBX594	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice,	Home&KitchenKitchen&Hom eAppliancesSmallKitchenAppl iancesHandBlenders	669.0	1499.0	0.55
3	B0B7LB6YCB	Green Tales Heat Seal Mini Food Sealer-Impulse Machine for Sealing Plastic Bags Packaging	Home&KitchenKitchen&Hom eAppliancesSmallKitchenAppl iancesVacuumSealers	161.0	300.0	0.46
4	B0BNDGL26T	MR. BRAND Portable USB Juicer Electric USB Juice	Home&KitchenKitchen&Hom eAppliancesSmallKitchenAppl	499.0	2199.0	0.77

페이지당 결과 수: 50 1 ~ 50 (전체 1400행)

- product_id, product_name 중복 여부: 같은 id, name을 가진 상품 총 45개(각각 2~3개 존재함)

```
SELECT product_id, product_name, COUNT(*) as cnt
FROM `modulabs_project.amazon`
GROUP BY product_id, product_name
HAVING cnt > 1
```

행	product_id	product_name	cnt
1	B0B5B6PQCT	boat Wave Call Smart Watch, Smart Talk with Advanced Dedicated Bluetooth Calling Chip, 1.69" HD Display with 550 NITS & 70% Color Gamut,	2
2	B0B6MSW6CT	Source Fast Phone Charging Cable & Data Sync USB Cable Compatible for iPhone 13, 12,11, X, 8, 7, 6, 5, iPad Air, Pro, Mini & iOS Devices	3
3	B08BCKN299	Source Gold Plated 3.5 mm Headphone Splitter for Computer 2 Male to 1 Female 3.5mm Headphone Mic Audio Y Splitter Cable Smartphone	2
4	B0VY1XK9NT	boat Wave Lite Smartwatch	2

페이지당 결과 수: 50 1 ~ 45 (전체 459)

- B0B5B6PQCT: 같은 상품인데 시기에 따라 할인율이 다르고 리뷰 개수가 다름 → 리뷰 개수가 더 많은 상품이 더 최신 데이터라고 가정하고 선택하여 업데이트

행	product_id	product_name	category	discounted_price	actual_price	discount_percent	rating	rating_count	about_product
1	B0B5B6PQCT	boat Wave Call Smart Watch, Smart Talk with Advanced Dedicated Bluetooth Calling Chip, 1.69" HD Display with 550 NITS & 70% Color Gamut,	ElectronicsWearableTechnol...	1799.0	7990.0	0.77	3.8	1788	Bluetooth Calling Wave Call comes with a premium built-in speaker and bluetooth calling via which you can stay connected to your friends.
2	B0B5B6PQCT	boat Wave Call Smart Watch, Smart Talk with Advanced Dedicated Bluetooth Calling Chip, 1.69" HD Display with 550 NITS & 70% Color Gamut,	ElectronicsWearableTechnol...	1999.0	7990.0	0.75	3.8	1781	Bluetooth Calling Wave Call comes with a premium built-in speaker and bluetooth calling via which you can stay connected to your friends.

- B0B6MSW6CT: 리뷰 개수가 같은데 원가가 다른 데이터 오류거나 시기가 다를 가능성이 있어보임

행	product_id	product_name	category	discounted_price	actual_price	discount_percent	rating	rating_count	about_product
1	B0B6MSW6CT	Source Fast Phone Charging Cable & Data Sync USB Cable Compatible for iPhone 13, 12,11, X, 8, 7, 6, 5, iPad Air, Pro, Mini & iOS Devices	ComputersAccessoriesPhone AccessoriesPeripheralsCables& AccessoriesCablesUSB Cables	199.0	999.0	0.8	3.9	7928	[Fast Charge& Data Sync] With built-in safety protections and four-core copper wires promote maximum signal quality and
2	B0B6MSW6CT	Source Fast Phone Charging Cable & Data Sync USB Cable Compatible for iPhone 13, 12,11, X, 8, 7, 6, 5, iPad Air, Pro, Mini & iOS Devices	ComputersAccessoriesPhone AccessoriesPeripheralsCables& AccessoriesCablesUSB Cables	199.0	999.0	0.8	3.9	7928	[Fast Charge& Data Sync] With built-in safety protections and four-core copper wires promote maximum signal quality and
3	B0B6MSW6CT	Source Fast Phone Charging Cable & Data Sync USB Cable Compatible for iPhone 13, 12,11, X, 8, 7, 6, 5, iPad Air, Pro, Mini & iOS Devices	ComputersAccessoriesPhone AccessoriesPeripheralsCables& AccessoriesCablesUSB Cables	199.0	1899.0	0.9	3.9	7928	[Fast Charge& Data Sync] With built-in safety protections and four-core copper wires promote maximum signal quality and

- B08BCKN299: 논으로 확인했을 때, 완전 동일해보임

행	product_id	product_name	category	discounted_price	actual_price	discount_percent	rating	rating_count	about_product
1	B08BCKN299	Source Gold Plated 3.5 mm Headphone Splitter for Computer 2 Male to 1 Female 3.5mm Headphone Mic Audio Y Splitter Cable Smartphone	ElectronicsHeadphonesEarbud...	120.0	999.0	0.88	3.9	6491	WIDE-RANGING USE: The cable features one 3.5mm Female and two 3.5mm Male connectors, enabling you to connect a 4-position headset
2	B08BCKN299	Source Gold Plated 3.5 mm Headphone Splitter for Computer 2 Male to 1 Female 3.5mm Headphone Mic Audio Y Splitter Cable Smartphone	ElectronicsHeadphonesEarbud...	120.0	999.0	0.88	3.9	6491	WIDE-RANGING USE: The cable features one 3.5mm Female and two 3.5mm Male connectors, enabling you to connect a 4-position headset

행	product_id	cnt	name_cnt	category_cnt	discounted_price...	actual_price_cnt	discount_percent...	rating_cnt	rating_count_cnt
1	B08BCKN299	2		1	1	1		1	1

about_product_cnt	user_id_cnt	user_name_cnt	review_id_cnt	review_title_cnt	review_content_cnt
1	1	1	1	1	2

- 3가지 상품을 뽑아서 직접 중복 데이터 확인한 후 **중복 데이터 정리 3가지 기준** 정의
 - 같은 상품 중에서 **리뷰 수가 더 많은 상품**이 최신 데이터일 확률이 높기 때문에 선택
 - 리뷰 수가 같으면 **할인율이 큰 상품**으로 소비자 입장에서 더 싸게 살 수 있는 상품으로 선택
 - 할인율도 같다면 **평점이 더 높은 상품**으로 선택
- 중복 상품 데이터 제거 후 테이블 업데이트(1400행 → 1351행)

```
CREATE OR REPLACE TABLE `modulabs_project.amazon` AS
SELECT *
EXCEPT(rn)
FROM (
  SELECT *,
    ROW_NUMBER() OVER (
      PARTITION BY product_id
      ORDER BY rating_count DESC, discount_percentage DESC, rating DESC
    ) AS rn
  FROM `modulabs_project.amazon`
)
WHERE rn = 1
```

행	product_id	product_name	category	discounted_price	actual_price	discount_percent...
1	B0BPJ8TB3F	Khaltan ORFin Fan heater for H...	Home&Kitchen Heating,Coolin... g&AirQuality RoomHeaters Fa... nHeaters	1299.0	2495.0	0.48
2	B0BFBNXS94	Personal Size Blender, Portable Blender, Battery Powered USB Blender, with Four Blades, Mini Blender Travel Bottle for Juice,	Home&Kitchen Kitchen&Hom... eAppliances SmallKitchenAppl... iances HandBlenders	669.0	1499.0	0.55
3	B0B7L86YCB	Green Tales Heat Seal Mini Food Sealer-Impulse Machine for Sealing Plastic Bags Packaging	Home&Kitchen Kitchen&Hom... eAppliances SmallKitchenAppl... iances VacuumSealers	161.0	300.0	0.46

- rating** : **STRING** 타입으로 되어있음
- **I** 들어가는 1행 삭제하고 **FLOAT64** 타입으로 수정 (1351행 → 1350행)

행	rating	cnt
15	3.8	84
16	3.9	114
17	4	52
18	4.0	107
19	4.1	225
20	4.2	207
21	4.3	209
22	4.4	114
23	4.5	68
24	4.6	16
25	4.7	6
26	4.8	3
27	5.0	3
28	I ?	1

```
# 이렇게 변경해주면 해당 rating 컬럼 테이블에서 제일 마지막에 들어감
# FLOAT로 하면 에러남 FLOAT64로 변경해줘야함
CREATE OR REPLACE TABLE `modulabs_project.amazon` AS
SELECT
```

```
* EXCEPT(rating),
  CAST(rating AS FLOAT64) AS rating
FROM `modulabs_project.amazon`
```

```
# 원하는 위치에 넣으려면 전체 컬럼 순서에 맞춰서 넣어줘야함
CREATE OR REPLACE TABLE `modulabs_project.amazon` AS
SELECT
    product_id,
    product_name,
    category,
    discounted_price,
    actual_price,
    discount_percentage,
    CAST(rating AS FLOAT64) AS rating,
    rating_count,
    about_product,
    user_id,
    user_name,
    review_id,
    review_title,
    review_content
FROM `modulabs_project.amazon`
```

☐	필드 이름	유형	모드
☐	product_id	STRING	NULLABLE
☐	product_name	STRING	NULLABLE
☐	category	STRING	NULLABLE
☐	discounted_price	FLOAT	NULLABLE
☐	actual_price	FLOAT	NULLABLE
☐	discount_percentage	FLOAT	NULLABLE
☐	rating	FLOAT	NULLABLE
☐	rating_count	INTEGER	NULLABLE
☐	about_product	STRING	NULLABLE
☐	user_id	STRING	NULLABLE
☐	user_name	STRING	NULLABLE
☐	review_id	STRING	NULLABLE
☐	review_title	STRING	NULLABLE
☐	review_content	STRING	NULLABLE

데이터 EDA

- 고유한 `product_id` 가 몇 개인가? **1350개**
- 고유한 `product_name` 은 몇 개인가? **1336개**
- `product_id` 컬럼을 제외 `product_name` 뿐 아니라 전체 컬럼이 완전 동일한 제품인지 확인

→ **product_id** 는 다르고 나머지 컬럼 완전히 동일 1개짜리 1개, 2개짜리 5개 총 6개 상품 그룹에서 중복 발생 → id를 제외한 모든 컬럼이 동일한 상품 중복 제거(1350행 → 1343행)

[illegible]

- 고유한 `product id` 가 몇 개인가? **1343개**

1. Firestick Remote : 2개 → 다른 상품으로 확인 그대로 유지

작업 항목	결과	시각화	JSON	생성 세부 정보	실용성						
	infected_id		product_name	category	discounted_price	actual_price	discount_percent	rating	rating_count	about_product	user_id
1	809714M9G3		FreeStick Remote	Electronics>HomeTheaters>TV&V...	1280.0	2495.0	0.48	3.3	75	by IDEE Compatible for Amazon Fire TV Stick (3rd Gen, 2021) Compatible for Amazon Fire TV Stick Lite Compatible for Amazon Fire TV Stick 4K Compatible for Amazon Fire TV Stick 4K Max	AH4747Q2P0BMM4HEEM BAZ0M26M4JZYV4V5202D G0P2C0R0N9KL4AETZ7D XW40C02E0260C026L2C JLAH8V6M06F6K4Z02N2
2	B0B8C20Z4H		FreeStick Remote	Electronics>HomeTheaters>TV&V...	1434.0	3995.0	0.64	4.0	32	Compatible for Amazon Fire TV Stick (3rd Gen, 2021) Compatible for Amazon Fire TV Stick Lite Compatible for Amazon Fire TV Stick 4K Compatible for Amazon Fire TV Stick 4K Max	ASCP4P0K000000276XU0 2ST07Q4AD0H9F7AD29E H0N1ALJ4K0Q4F04H05E7 J0Y0VM40060000003V5GAA F0K0ZJC8B4C0F0C0R0P0U0E

- | product_id | product_name | discount_price | actual_price | discount_percent | rating | review_count | about_product |
|------------|--------------|----------------|--------------|------------------|--------|--------------|---------------|
| 1 | 8099FMMATC | | | | | | |
| 2 | 8099F343W | | | | | | |
| 3 | 8099F343W | | | | | | |
| 4 | 8099F343W | | | | | | |
| 5 | 8099F343W | | | | | | |
| 6 | 8099F343W | | | | | | |
| 7 | 8099F343W | | | | | | |
| 8 | 8099F343W | | | | | | |
| 9 | 8099F343W | | | | | | |
| 10 | 8099F343W | | | | | | |
| 11 | 8099F343W | | | | | | |
| 12 | 8099F343W | | | | | | |
| 13 | 8099F343W | | | | | | |
| 14 | 8099F343W | | | | | | |
| 15 | 8099F343W | | | | | | |
| 16 | 8099F343W | | | | | | |
| 17 | 8099F343W | | | | | | |
| 18 | 8099F343W | | | | | | |
| 19 | 8099F343W | | | | | | |
| 20 | 8099F343W | | | | | | |
| 21 | 8099F343W | | | | | | |
| 22 | 8099F343W | | | | | | |
| 23 | 8099F343W | | | | | | |
| 24 | 8099F343W | | | | | | |
| 25 | 8099F343W | | | | | | |
| 26 | 8099F343W | | | | | | |
| 27 | 8099F343W | | | | | | |
| 28 | 8099F343W | | | | | | |
| 29 | 8099F343W | | | | | | |
| 30 | 8099F343W | | | | | | |
| 31 | 8099F343W | | | | | | |
| 32 | 8099F343W | | | | | | |
| 33 | 8099F343W | | | | | | |
| 34 | 8099F343W | | | | | | |
| 35 | 8099F343W | | | | | | |
| 36 | 8099F343W | | | | | | |
| 37 | 8099F343W | | | | | | |
| 38 | 8099F343W | | | | | | |
| 39 | 8099F343W | | | | | | |
| 40 | 8099F343W | | | | | | |
| 41 | 8099F343W | | | | | | |
| 42 | 8099F343W | | | | | | |
| 43 | 8099F343W | | | | | | |
| 44 | 8099F343W | | | | | | |
| 45 | 8099F343W | | | | | | |
| 46 | 8099F343W | | | | | | |
| 47 | 8099F343W | | | | | | |
| 48 | 8099F343W | | | | | | |
| 49 | 8099F343W | | | | | | |
| 50 | 8099F343W | | | | | | |
| 51 | 8099F343W | | | | | | |
| 52 | 8099F343W | | | | | | |
| 53 | 8099F343W | | | | | | |
| 54 | 8099F343W | | | | | | |
| 55 | 8099F343W | | | | | | |
| 56 | 8099F343W | | | | | | |
| 57 | 8099F343W | | | | | | |
| 58 | 8099F343W | | | | | | |
| 59 | 8099F343W | | | | | | |
| 60 | 8099F343W | | | | | | |
| 61 | 8099F343W | | | | | | |
| 62 | 8099F343W | | | | | | |
| 63 | 8099F343W | | | | | | |
| 64 | 8099F343W | | | | | | |
| 65 | 8099F343W | | | | | | |
| 66 | 8099F343W | | | | | | |
| 67 | 8099F343W | | | | | | |
| 68 | 8099F343W | | | | | | |
| 69 | 8099F343W | | | | | | |
| 70 | 8099F343W | | | | | | |
| 71 | 8099F343W | | | | | | |
| 72 | 8099F343W | | | | | | |
| 73 | 8099F343W | | | | | | |
| 74 | 8099F343W | | | | | | |
| 75 | 8099F343W | | | | | | |
| 76 | 8099F343W | | | | | | |
| 77 | 8099F343W | | | | | | |
| 78 | 8099F343W | | | | | | |
| 79 | 8099F343W | | | | | | |

- | # | product_id | product_name | category | discount_price | actual_price | discount_percent | rating | rating_count | about_product |
|---|------------|---|---------------------------------|----------------|--------------|------------------|--------|--------------|--|
| 1 | D08F44VFX | Fire-Bolt Nrga Cat Pro Plus
1.8T Smart Watch with
Bluetooth Calling, A/V Voice
Assistant, 100 Sports
Modes (IP67 Rating, 240x230) | Electronics/Wearable Technology | 1799.0 | 1999.0 | 0.91 | 4.2 | 12037 | Fire-Bolt is India's No. 1 Wearable Watch Brand Q122 by IDC Worldwide quarterly wearable device tracker
Q122 Bluetooth Calling Watch Fire-Bolt Nrga Cat Pro Plus Smart watch enables you to make and receive calls directly from your watch via the built-in speaker and microphone |
| 2 | D08F44VFX | Fire-Bolt Nrga Cat Pro Plus
1.8T Smart Watch with
Bluetooth Calling, A/V Voice
Assistant, 100 Sports
Modes (IP67 Rating, 240x230) | Electronics/Wearable Technology | 1799.0 | 1999.0 | 0.91 | 4.2 | 13937 | Fire-Bolt is India's No. 1 Wearable Watch Brand Q122 by IDC Worldwide quarterly wearable device tracker
Q122 Bluetooth Calling Watch Fire-Bolt Nrga Cat Pro Plus Smart watch enables you to make and receive calls directly from your watch via the built-in speaker and microphone |
| 3 | D08F44VFX | Fire-Bolt Nrga Cat Pro Plus
1.8T Smart Watch with
Bluetooth Calling, A/V Voice
Assistant, 100 Sports
Modes (IP67 Rating, 240x230) | Electronics/Wearable Technology | 1799.0 | 1999.0 | 0.91 | 4.2 | 13937 | Fire-Bolt is India's No. 1 Wearable Watch Brand Q122 by IDC Worldwide quarterly wearable device tracker
Q122 Bluetooth Calling Watch Fire-Bolt Nrga Cat Pro Plus Smart watch enables you to make and receive calls directly from your watch via the built-in speaker and microphone |

- | | product_id | product_name | category | discount_price | stock_price | discount_percent | rating | rating_count | about_product |
|---|------------|--|-------------------------------------|----------------|-------------|------------------|--------|--------------|---|
| 1 | B0B3BRWFG6 | Free-Bolt Phoenix Smart Watch with Bluetooth Calling
1.31"120° Sports Mode
240x240 PPI High Res with IP68, Heart Rate Monitoring & GPS | Electronics/ wearable Technology... | 199.0 | 999.0 | 0.0 | 4.5 | 2769 | Free-Bolt is India No 1 Wearable Watch Brand Q1'22 by GC Worldwide quarterly wearable device tracker
Q1'22 : Bluetooth Calling Watch No 1 - Free-Bolt Phoenix enables you to make and receive calls directly from your watch via the built-in speaker and microphone. This smartwatch features the |
| 2 | B0B3B5D2G3 | Free-Bolt Phoenix Smart Watch with Bluetooth Calling
1.31"120° Sports Mode
240x240 PPI High Res with IP68, Heart Rate Monitoring & GPS | Electronics/ wearable Technology... | 199.0 | 999.0 | 0.0 | 4.3 | 2786 | Free-Bolt is India No 1 Wearable Watch Brand Q1'22 by GC Worldwide quarterly wearable device tracker
Q1'22 : Bluetooth Calling Watch No 1 - Free-Bolt Phoenix enables you to make and receive calls directly from your watch via the built-in speaker and microphone. This smartwatch features the |
| 3 | B0B3B5D2GF | Free-Bolt Phoenix Smart Watch with Bluetooth Calling
1.31"120° Sports Mode
240x240 PPI High Res with IP68, Heart Rate Monitoring & GPS | Electronics/ wearable Technology... | 199.0 | 999.0 | 0.0 | 4.3 | 2784 | Free-Bolt is India No 1 Wearable Watch Brand Q1'22 by GC Worldwide quarterly wearable device tracker
Q1'22 : Bluetooth Calling Watch No 1 - Free-Bolt Phoenix enables you to make and receive calls directly from your watch via the built-in speaker and microphone. This smartwatch features the |

- | | product name | category | discount price | actual price | discount percent | ratio | ratio count | about count |
|---|--------------|--|---------------------------------|--------------|------------------|-------|-------------|-------------|
| 1 | SEB02PC29 | Fire-Built Version 1.70
AMOLED Bratwatch Calling
Smartwatch with 360*440
Pixel Resolution 1050 Sports
Mode, Two Connection, Voice
AMOLED Version 1.70 | Electronics/Wearable/Technology | 3999.0 | 1799.0 | 0.78 | 4 | 37181 |
| 2 | SEB02PC30A | AMOLED Bratwatch Calling
Smartwatch with 360*440
Pixel Resolution 1050 Sports
Mode, Two Connection, Voice
AMOLED Version 1.70 | Electronics/Wearable/Technology | 3999.0 | 1699.0 | 0.76 | 4 | 17762 |

- | 행 | product_name | cnt |
|---|------------------|-----|
| 1 | Firestick Remote | 2 |

- ## SQL Project2(아마존 판매 데이터셋 최홍주)

행	rating	cnt
3	2.6	1
4	2.8	2
5	2.9	1
6	3.0	4
7	3.1	4
8	3.2	2
9	3.3	15
10	3.4	10
11	3.5	26
12	3.6	34
13	3.7	41
14	3.8	84
15	3.9	114
16	4.0	159
17	4.1	221
18	4.2	201
19	4.3	206
20	4.4	114
21	4.5	68
22	4.6	16
23	4.7	6
24	4.8	3
25	5.0	3

- 평점 별 평균 리뷰 수는? 5점짜리 상품은 3개에 리뷰 수도 굉장히 적음

행	rating	avg_reviews	cut
1	5.0	9.666666666666...	3
2	4.8	19265.0	3
3	4.7	3398.0	6
4	4.6	7447.875	16
5	4.5	17726.05882352...	68
6	4.4	25633.05263157...	114
7	4.3	16312.27184466...	206
8	4.2	20551.76119402...	201
9	4.1	29579.27149321...	221
10	4.0	13338.08805031...	159
11	3.9	13453.46491228...	114
12	3.8	10087.88095238...	84
13	3.7	5046.780487804...	41
14	3.6	5103.147058823...	34
15	3.5	8208.5	26
16	3.4	3275.4	10
17	3.3	2759.066666666...	15
18	3.2	199.0	2
19	3.1	1522.25	4
20	3.0	350.5	4
21	2.9	159.0	1
22	2.8	98.0	2
23	2.6	24.0	1

- 카테고리들 '|' 기준으로 cat_1, cat_2, cat_3까지 나눴을 때, cat_1의 분포는? 총 9종류

행	cat_1	cut
1	Electronics	477
2	Home&Kitchen	447
3	Computers&Accessories	375
4	OfficeProducts	31
5	MusicalInstruments	2
6	HomeImprovement	2
7	Car&Motorbike	1
8	Health&PersonalCare	1
9	Toys&Games	1

- cat_2 에는 어떤 상품들이 있는가? 총 29종류

행	cat_2	cut
1	Accessories&Peripherals	307
2	Kitchen&HomeAppliances	307
3	HomeTheater,TV&Video	155
4	Mobiles&Accessories	147
5	Heating,Cooling&AirQuality	116
6	Headphones,Earbuds&Accessor...	63
7	WearableTechnology	54
8	NetworkingDevices	30
9	OfficePaperProducts	27
10	ExternalDevices&DataStorage	18
11	HomeStorage&Organization	16
12	HomeAudio	16
13	Cameras&Photography	16
14	GeneralPurposeBatteries&Batte...	14
15	Accessories	11
16	Printers,Inks&Accessories	11
17	CraftMaterials	7
18	Components	5
19	OfficeElectronics	4
20	Electrical	2

- cat_1(Electronics) 하위 카테고리 분포

```
SELECT cat_2, COUNT(*) AS cut
FROM `modulabs_project.amazon_cat_split`
WHERE cat_1 = 'Electronics'
GROUP BY cat_2
```

행	cat_2	cut
1	HomeTheater,TV&Video	155
2	Mobiles&Accessories	147
3	Headphones,Earbuds&Accessor...	63
4	WearableTechnology	54
5	HomeAudio	16
6	Cameras&Photography	16
7	GeneralPurposeBatteries&Batte...	14
8	Accessories	11
9	PowerAccessories	1

- cat_1(Home&Kitchen) 하위 카테고리 분포

행	cat_2	cut
1	Kitchen&HomeAppliances	307
2	Heating,Cooling&AirQuality	116
3	HomeStorage&Organization	16
4	CraftMaterials	7
5	Kitchen&Dining	1

- cat_1(Computers&Accessories) 하위 카테고리 분포

행	cat_2	cut
1	Accessories&Peripherals	307
2	NetworkingDevices	30
3	ExternalDevices&DataStorage	18
4	Printers,Inks&Accessories	11
5	Components	5
6	Monitors	2
7	Laptops	1
8	Tablets	1

- cat_3 에는 어떤 상품들이 있는가? 총 72종류

행	cat_3	cut
1	SmallKitchenAppliances	181
2	Cables&Accessories	167
3	Accessories	93
4	Vacuum,Cleaning&Ironing	81
5	MobileAccessories	74
6	Smartphones&BasicMobiles	73
7	Televisions	66
8	Headphones	60
9	Keyboards,Mice&InputDevices	60
10	SmartWatches	54
11	RoomHeaters	46
12	WaterHeaters&Geysers	44
13	LaptopAccessories	33
14	Paper	27
15	WaterPurifiers&Accessories	26
16	Fans	17
17	NetworkAdapters	16
18	LaundryOrganization	16
19	Coffee,Tea&Espresso	15
20	PCGamingPeripherals	15

- cat_3 기준으로 카운트 세고 cat_1, cat_2 함께 확인

행	cat_1	cat_2	cat_3	cut
1	Home&Kitchen	Kitchen&HomeAppliances	SmallKitchenAppliances	181
2	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	167
3	Electronics	HomeTheater,TV&Video	Accessories	82
4	Home&Kitchen	Kitchen&HomeAppliances	Vacuum,Cleaning&Ironing	81
5	Electronics	Mobiles&Accessories	MobileAccessories	74
6	Electronics	Mobiles&Accessories	Smartphones&BasicMobiles	73
7	Electronics	HomeTheater,TV&Video	Televisions	66
8	Computers&Accessories	Accessories&Peripherals	Keyboards,Mice&InputDevices	60
9	Electronics	Headphones,Earbuds&Accessor...	Headphones	60
10	Electronics	WearableTechnology	SmartWatches	54
11	Home&Kitchen	Heating,Cooling&AirQuality	RoomHeaters	46
12	Home&Kitchen	Heating,Cooling&AirQuality	WaterHeaters&Geysers	44
13	Computers&Accessories	Accessories&Peripherals	LaptopAccessories	33
14	OfficeProducts	OfficePaperProducts	Paper	27
15	Home&Kitchen	Kitchen&HomeAppliances	WaterPurifiers&Accessories	26
16	Home&Kitchen	Heating,Cooling&AirQuality	Fans	17
17	Computers&Accessories	NetworkingDevices	NetworkAdapters	16
18	Home&Kitchen	HomeStorage&Organization	LaundryOrganization	16
19	Home&Kitchen	Kitchen&HomeAppliances	Coffee,Tea&Espresso	15
20	Computers&Accessories	Accessories&Peripherals	PCGamingPeripherals	15

- 같은 소분류(cat_3) 안에서 MIN , MAX , AVG 은 얼마인지? 전체적으로 카테고리별 가장 싼 상품과 가장 비싼 상품의 가격 차이가 많이 남

행	cat_1	cat_2	cat_3	product_count	min_price	max_price	avg_price
1	Home&Kitchen	Kitchen&HomeAppliances	SmallKitchenAppliances	181	79.0	23999.0	3200.8
2	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	167	199.0	2100.0	942.07
3	Electronics	HomeTheater,TV&Video	Accessories	82	399.0	4999.0	1273.9
4	Home&Kitchen	Kitchen&HomeAppliances	Vacuum,Cleaning&Ironing	81	299.0	5990.0	4279.38
5	Electronics	Mobiles&Accessories	MobileAccessories	74	171.0	6999.0	1688.26
6	Electronics	Mobiles&Accessories	Smartphones&BasicMobiles	73	1249.0	74999.0	18311.75
7	Electronics	HomeTheater,TV&Video	Televisions	66	11000.0	19990.0	38643.3
8	Computers&Accessories	Accessories&Peripherals	Keyboards,Mice&InputDevices	60	237.0	4699.0	1265.13
9	Electronics	Headphones,Earbuds&Accessor...	Headphones	60	399.0	15990.0	2976.42
10	Electronics	WearableTechnology	SmartWatches	54	999.0	29999.0	7578.31

추천 시스템 1

1. 추천 시스템 이름

→ “놓치면 후회하는 초특가 득템 상품 추천” (할인율이 높은 가성비 & 고평점 결합 추천)

2. 추천 시스템의 테마

- 단순히 할인율만 높으면 안 팔려서 할인 하는 제품일 수 있다고 생각하여, 리뷰 수와 평점으로 필터링하여 ‘진짜 좋은데 싸게 파는 제품’을 추천
- 일정 수준 이상의 검증된 퀄리티를 갖춘 상품 중 할인율이 가장 높은 상품 추천

3. 구현 로직

→ 평점 4.0 이상, 리뷰 수 100개 이상, 50% 이상의 할인율 조건을 걸고 할인율이 높은 순으로 정렬하고 평점도 높은 순으로 정렬했습니다.

```
SELECT
    product_id,
    product_name,
    discounted_price,
    actual_price,
    discount_percentage,
    rating,
    rating_count,
    cat_1,
    cat_2,
    cat_3
FROM `modulabs_project.amazon_cat_split`
WHERE rating >= 4.0
    AND rating_count >= 100
    AND discount_percentage >= 0.5
ORDER BY discount_percentage DESC, rating DESC
LIMIT 5
```

4. 결과

→ 1위로 추천된 USB C타입 어댑터의 경우, 94%의 높은 할인율을 보이면서도 4,400개가 넘는 리뷰와 4.3점의 우수한 평점을 보유하고 있습니다. 2위인 스마트워치(Fire-Boltt) 역시 13,000개가 넘는 압도적인 리뷰 수와 4.2점의 평점을 기록한 인기 상품임에도 무려 91%의 할인이 적용되었습니다. 결과적으로 “놓치면 후회하는 초특가 득템”이라는 시스템의 기획 의도에 맞게, 품질이 검증된 파격 할인 상품들이 성공적으로 필터링 된 것을 확인할 수 있었습니다.

→ 비즈니스 기대 효과 : 단순히 안 팔리는 악성 재고를 할인 기준으로만 묶어서 보여주는 것이 아니라, “많은 사람들이 이미 만족한 인기 상품을 현재 엄청 싸게 살 수 있다”는 점을 어필함으로써 고객의 구매 전환율을 올릴 수 있습니다.

랭	product_id	product_name	discounted_price	actual_price	discount_percentage	rating	rating_count	cat_1	cat_2	cat_3
1	B071C2M40C	Its 3 Pin Mini USB C Type C Adapter Plug Type C Female to USB A Male Charger Charging Cable Adapter Converter compatible for	294.0	4999.0	0.94	4.3	4426	Computers&Accessories	Accessories&Peripherals	Adapters
2	B0BF54LX95	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch with Bluetooth Calling AI Voice Assistance, 100 Sports Modes IP67 Rating, 1407000	1799.0	19999.0	0.91	4.2	13937	Electronics	WearableTechnology	SmartWatches
3	B0BBW521YC	LAPSTER 12pcs Spiral Cable Protectors for Charger, Wires, Data Charger Cable Protector for Computers, Cell Phones etc.(Grey)	99.0	999.0	0.9	4.4	305	Electronics	Mobiles&Accessories	MobileAccessories
4	B0BWS61K93	LAPSTER Spiral Charger Spiral Charger Cable Protectors for Wires Data Cable Saver Charging Cord Protective Cable Cover Set of 3 (12	99.0	999.0	0.9	4.1	8751	Computers&Accessories	Accessories&Peripherals	Cables&Accessories
5	B095J1FF1Y	Source Protective Case Cover Compatible Boat Xland Overall Protective Case TPU HD Clear	199.0	1899.0	0.9	4.0	4740	Electronics	Mobiles&Accessories	MobileAccessories

추천 시스템 2

1. 추천 시스템 이름

→ “카테고리별 베스트 상품 추천” (세분화된 카테고리 랭킹 추천)

2. 추천 시스템의 테마

→ **cat_3** (소분류)별 상위 1~3위(Top 3) 상품을 뽑아내서 추천

3. 구현 로직

→ **cat_3** (소분류)별 상품 수가 가장 많은 핵심 카테고리 5개에서 평점과 리뷰 수를 기준으로 순위 부여하고 최소 리뷰 수 50개 이상 기준을 추가했습니다.

```
WITH RankedProducts AS (
    SELECT
        cat_1,
        cat_2,
        cat_3,
        product_id,
        product_name,
        rating,
        rating_count,
        ROW_NUMBER() OVER(PARTITION BY cat_3 ORDER BY rating DESC, rating_count DESC) as rnk
    FROM `modulabs_project.amazon_cat_split`
    WHERE
        cat_3 IN (
```

```

        'SmallKitchenAppliances',
        'Cables&Accessories',
        'Accessories',
        'Vacuum,Cleaning&Ironing',
        'MobileAccessories'
    )
    AND rating_count >= 50
)
SELECT
    cat_1,
    cat_2,
    cat_3,
    rn AS category_rank,
    product_id,
    product_name,
    rating,
    rating_count
FROM RankedProducts
WHERE rn <= 3
ORDER BY cat_1, cat_2, cat_3, rn

```

4. 결과

→ cat_1 (대분류)부터 cat_3 (소분류)까지 함께 나열하여, 고객이 현재 어떤 카테고리의 랭킹을 보고 있는지 직관적으로 파악할 수 있도록 가독성을 높이고, 'Cables&Accessories' 카테고리에서는 무려 10만 개가 넘는 압도적인 리뷰 수를 가진 'AmazonBasics USB 케이블' 시리즈가 1~3위를 휩쓸며 해당 분야에서의 베스트 상품임을 입증하고, 'Accessories' 소분류라는 이름이라도 상위 카테고리에 따라 카메라용 필름과 홈시어터용 HDMI 케이블로 명확히 나뉘어 각각의 베스트셀러가 정확히 추천되는 것을 확인할 수 있었습니다.

→ 비즈니스 기대 효과 : 고객이 특정 목적을 가지고 카테고리에 진입했을 때, 가장 대중적으로 검증된 상품을 1위부터 3위까지 추천해줌으로써 탐색 피로도를 줄이고 빠른 구매 결정을 유도할 수 있습니다.

rank	cat_1	cat_2	cat_3	category_rank	product_id	product_name	rating	rating_count
1	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	1	B0DNH11K1K	AmazonBasics USB 2.0 Cable - A-Male to B-Male - for Personal Computer, Printer- 6 Feet (1.8 Meters), Black	4.5	107687
2	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	2	B0DNH13QBW	AmazonBasics USB 2.0 Extension Cable for Personal Computers, Printer, 2-Pack - A-Male to A-Female - 3.3 Feet (1 Meter), Black	4.5	74977
3	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	3	B0DNH11PEY	AmazonBasics USB 2.0 - A-Male to A-Female Extension Cable for Personal Computer, Printer (Black, 9.8 Feet/3 Meters)	4.5	74976
4	Electronics	Cameras&Photography	Accessories	2	B0DR1P3BA0	Fujifilm Instax Mini Single Pack 10 Sheets Instant Film for Fujifilm Instant Cameras	4.5	4875
5	Electronics	HomeTheater,Tv&Video	Accessories	1	B08TG6316Z	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ultra High-Speed HDMI 2.1 Cable 4k 120Hz 144Hz 2k 165Hz 8k 60Hz Dynamic HDR ARC eARC	4.5	3664
6	Electronics	HomeTheater,Tv&Video	Accessories	3	B07RYZK9MP	AmazonBasics 10.2 Gbps High-Speed 4K HDMI Cable with Braided Cord (10-Foot)	4.5	1029

추천 시스템 3

1. 추천 시스템 이름

→ “이 상품을 본 고객들이 함께 찾은 상품 추천” (유저 기반 연관 추천)

2. 추천 시스템의 테마

→ A상품을 리뷰한 유저가 B상품도 리뷰했다는 동시 구매(리뷰) 패턴을 찾아 연관 상품 추천

3. 구현 로직

→ user_id 컬럼을 활용하여 , 기준으로 유저를 분리하여 동시 구매한 상품을 찾습니다.

```

WITH UnnestedUsers AS (
    SELECT
        product_id,
        product_name,
        cat_3,
        TRIM(user) AS user_id
    FROM `modulabs_project.amazon_cat_split`,
        UNNEST(SPLIT(user_id, ',')) AS user
    WHERE user != ''
)

SELECT

```

```

t1.product_id AS target_product_id,
t1.product_name AS target_product_name,
t1.cat_3 AS target_category,
t2.product_id AS recommended_product_id,
t2.product_name AS recommended_product_name,
COUNT(DISTINCT t1.user_id) AS co_purchase_count -- 동시 구매/리뷰 유저 수
FROM UnnestedUsers t1
JOIN UnnestedUsers t2
  ON t1.user_id = t2.user_id
  AND t1.product_id != t2.product_id -- 자기 자신은 제외
GROUP BY t1.product_id, t1.product_name, t1.cat_3, t2.product_id, t2.product_name
ORDER BY co_purchase_count DESC
LIMIT 10

```

4. 결과

→ 동일한 고객(**user_id**)이 함께 구매하거나 리뷰를 남긴 이력이 가장 많은 상품 함께를 추출했는데, **target_product_id** 와 **recommended_product_id** 가 다르지만 상품명은 거의 동일한 '색상/옵션/버전 차이' 상품들이 상위권을 차지했습니다. 아마존 특성상 색상이나 세부 스펙 모델이 다른 같은 상품들이 서로 다른 상품 ID를 가지고 있어 나타난 결과이며, 고객들이 스마트워치나 모바일 액세서리를 구매할 때에 동일 라인업의 다른 색상이나 세부 스펙 모델을 동시에 구매하거나 비교하는 데이터 특성을 확인할 수 있었습니다.

→ 비즈니스 기대 효과 : 고객이 특정 상품을 보고 있을 때, 다른 고객들이 함께 비교하거나 구매한 다른 옵션을 제시하여 고객이 원하는 정확한 옵션(색상, 사이즈 등)을 찾지 못해 이탈하는 것을 방지하고, 또는 “이 색상도 예쁘네 같이 살까?”하는 추가 구매를 유도할 수 있습니다.

행	target_product_id	target_product_name	target_category	recommended_product_id	recommended_product_name	co_purchase_count
1	B0BK4RDQ71	Tukzer Capacitive Stylus Pen for Touch Screens Devices, Fine Point, Lightweight Metal Body with Magnetism Cover Cap for	MobileAccessories	B0BK4PSZ3V	Tukzer Capacitive Stylus Pen for Touch Screens Devices, Fine Point, Lightweight Metal Body with Magnetism Cover Cap for	8
2	B0B53NXFFR	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full Touch Display, Real Heart Rate Monitor, SpO2, Watch Faces, 5	SmartWatches	B0B53QFZPY	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full Touch Color Display, Real Heart Rate Monitor, SpO2,	8
3	B0B5D0JNH4	boAt Wave Call Smart Watch, Smart Talk with Advanced Dedicated Bluetooth Calling Chip, 1.69" HD Display with 550 NITS & 70% Color Gamut,	SmartWatches	B0B5B6PQCT	boAt Wave Call Smart Watch, Smart Talk with Advanced Dedicated Bluetooth Calling Chip, 1.69" HD Display with 550 NITS & 70% Color Gamut,	8
4	B0B5D0JNH4	boAt Wave Call Smart Watch, Smart Talk with Advanced Dedicated Bluetooth Calling Chip, 1.69" HD Display with 550 NITS & 70% Color Gamut,	SmartWatches	B0B5D39BCD	boAt Wave Call Smart Watch, Smart Talk with Advanced Dedicated Bluetooth Calling Chip, 1.69" HD Display with 550 NITS & 70% Color Gamut,	8
5	B0B53NXFFR	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full Touch Display, Real Heart Rate Monitor, SpO2, Watch Faces, 5	SmartWatches	B0B53QLB9H	PTron Newly Launched Force X10 Bluetooth Calling Smartwatch with 1.7" Full Touch Color Display, Real Heart Rate Monitor, SpO2,	8

추천 시스템 4

1. 추천 시스템 이름

→ “리뷰가 증명하는 진짜 찐템 상품 추천” (신뢰도 점수 기반 추천)

2. 추천 시스템의 테마

→ 점 5.0인데 리뷰가 1개인 상품과, 별점 4.7인데 리뷰가 10,000개인 상품 중 어느 것이 더 신뢰할 수 있는가 당연 후자의 상품을 신뢰할 것 입니다. 이를 수학적으로 보정하여 진정한 인기 상품을 추천

→ 소수의 리뷰로 인한 높은 평점은 제거하고, 리뷰 수와 평점을 종합적으로 고려한 ‘신뢰도 가중치 점수(Weighted Score)’ 기반 추천

3. 구현 로직

→ 베이지안 평균(Bayesian Average) 또는 가중 평점(Weighted Rating)은 데이터셋이 작거나 평점이 불균형할 때, 단순 평균의 신뢰성 문제를 보완하기 위해 사용하는 통계적 평균 방식입니다. 사전 정보(전체 평균)를 바탕으로 데이터가 적은 항목의 평점을 보정하여 신뢰도를 높이는 것이 핵심입니다.

→ 리뷰 수가 적은 상품이 상위권에 랭크되는 '단순 평균의 오류'를 방지하기 위해, 실무 추천 시스템에서 활용되는 베이지안 평균(Bayesian Average) 기반의 가중 평점을 활용하여 파생 변수를 만들었습니다.

→ $(rating * rating_count) / (rating_count + \text{최소임계값} = 100)$ 연산을 통해 **trust_score** 라는 새로운 컬럼을 만들었습니다. 이 공식을 활용했을 때, **리뷰가 2개인 5점짜리 상품은 0.09점**이 나오고 **리뷰가 5000개인 4.8점짜리 상품은 4.7점**으로 평점이 거의 그대로 유지되며 신뢰도 점수가 높기 때문에 실제로 리뷰가 증명하는 찐템 상품이라고 생각할 수 있습니다.

```

WITH TrustScoreCalc AS (
  SELECT
    cat_3,
    product_id,
    product_name,
    rating,
    rating_count,

```

```

-- 신뢰도 점수(가중 평점) 공식
-- (실제 평점 * 실제 리뷰수) / (실제 리뷰수 + 100)
(rating * rating_count) / (rating_count + 100) AS trust_score
FROM `modulabs_project.amazon_cat_split`
WHERE rating_count IS NOT NULL
)
SELECT
    cat_3,
    product_id,
    product_name,
    rating,
    rating_count,
    ROUND(trust_score, 2) AS trust_score
FROM TrustScoreCalc
ORDER BY trust_score DESC
LIMIT 10

```

4. 결과

→ 1위(Water Heater)와 2위(Instant Pot) 상품은 원래 평점(**rating**)이 4.8점으로 동일합니다. 하지만 리뷰 수가 약 53,000개인 1위 상품은 신뢰도 점수(**trust_score**)가 4.79점으로 원래 평점을 거의 유지했고, 리뷰 수가 약 3,900개인 2위 상품은 4.68점으로 약간의 페널티가 적용된 것을 확인할 수 있었습니다.

→ 비즈니스 기대 효과 : 고객에게 객관적으로 가장 많이 검증된, 실패 확률이 적은 고품질 상품을 우선적으로 추천함으로써 아마존에 대한 고객의 신뢰도를 높이고 반품률을 낮출 수 있습니다.

랭	cat_3	product_id	product_name	rating	rating_count	trust_score
1	WaterHeaters&Geysers	B0BR4F878Q	Swiffer Instant Electric Water Heater Faucet Tap Home-Kitchen Instantaneous Water Heater Tank less for Tap, LED Electric Head Water	4.8	53803	4.79
2	SmallKitchenAppliances	B0BS3DS4TF	Instant Pot Air Fryer, Vortex 2QT, Touch Control Panel, 360° EvenCrisp™ Technology, Uses 95 % less Oil, 4-in-1 Appliance: Air Fry, Roast, Bake,	4.8	3964	4.68
3	MobileAccessories	B0B23LW7NV	Spigen EZ Fit Tempered Glass Screen Protector for iPhone 14 Pro Max - 2 Pack (Sensor Protection)	4.7	7779	4.54
4	Televisions	B09WN3SRC7	Sony Bravia 164 cm (65 inches) 4K Ultra HD Smart LED Google TV KD-65X74K (Black)	4.7	5935	4.62
5	PCGamingPeripherals	B01J1CFO5I	Redgear MP35 Speed-Type Ga...	4.6	33434	4.59
6	MobileAccessories	B09SRTJH1M	Spigen EZ Fit Tempered Glass Screen Protector Guard for iPhone 14/13/13 Pro - 2 Pack	4.6	26603	4.58

추천 시스템 5

1. 추천 시스템 이름

→ “나의 지갑 사정에 맞는 맞춤 제안 (프리미엄 vs 가성비) 추천”

2. 추천 시스템의 테마

→ 동일한 종류의 상품을 찾더라도 유저마다 예산이 다르기 때문에 동일한 소분류(**cat_3**) 내에서 가장 비싼 그룹의 프리미엄 상품 1개와 가장 저렴한 가성비 그룹의 예산 절약형 상품 1개를 함께 추천

3. 구현 로직

→ **cat_3** 내 상품들의 가격(**actual_price**)을 기준으로 고가, 중가, 저가 3가지 등급으로 나눴습니다.

→ 너무 적은 상품 수를 가진 카테고리는 제대로 나뉘지지 않을 가능성이 있어 **cat_3** 내 상품 수가 가장 많은 Top 3 카테고리 확인했습니다.

```

WITH PriceCategories AS (
    SELECT
        cat_1,
        cat_2,
        cat_3,
        product_id,
        product_name,
        actual_price,
        rating,
        rating_count,
        NTILE(3) OVER(PARTITION BY cat_3 ORDER BY actual_price DESC) AS price_tier
    FROM `modulabs_project.amazon_cat_split`
)

```

```

WHERE rating_count >= 50
    -- 상품 수가 충분히 확보된 핵심 카테고리 3개만 필터링
    AND cat_3 IN ('SmallKitchenAppliances', 'Cables&Accessories', 'Accessories')
),
TopPerTier AS (
    SELECT
        cat_1,
        cat_2,
        cat_3,
        price_tier,
        product_id,
        product_name,
        actual_price,
        rating,
        rating_count,
        ROW_NUMBER() OVER(PARTITION BY cat_3, price_tier ORDER BY rating DESC, rating_count DESC) as
rnk
    FROM PriceCategories
)

SELECT
    cat_1,
    cat_2,
    cat_3,
    CASE
        WHEN price_tier = 1 THEN '👑 프리미엄 (하이엔드)'
        WHEN price_tier = 3 THEN '💎 가성비 (실속형)'
    END AS recommendation_type,
    product_id,
    product_name,
    actual_price,
    rating,
    rating_count
FROM TopPerTier
WHERE rnk = 1
    AND price_tier IN (1, 3)
ORDER BY cat_1, cat_2, cat_3, price_tier

```

4. 결과

→ 소형 주방가전(**SmallKitchenAppliances**) 카테고리를 보면, 프리미엄 라인으로는 고가 하이엔드 기기인 인스턴트 팻(Instant Pot)이, 가성비 라인으로는 실속형 계란 찜기(Egg Boiler)가 나란히 추천되었습니다. 같은 카테고리 내에서 예산 차이를 명확하게 확인 할 수 있었습니다.

→ 비즈니스 기대 효과 : 고객이 특정 카테고리에 진입했을 때, 가장 비싸고 좋은 모델 vs 가장 가성비 좋은 모델을 동시에 추천받음으로써 가격 장벽으로 인한 고객 이탈을 막고 구매 전환율을 높일 수 있습니다.

rank	cat_1	cat_2	cat_3	recommendation_type	product_id	product_name	actual_price	rating	rating_count
1	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	👑 프리미엄 (하이엔드)	B002762805	AmazonBasics New Release Nylon USB-A to Lightning Cable Cord, MFi Certified Charger for Apple iPhones, iPad, Smartwatch	2100.0	4.5	5492
2	Computers&Accessories	Accessories&Peripherals	Cables&Accessories	💎 가성비 (실속형)	B00N411K0K	AmazonBasics USB 2.0 Cable - A-Male to B-Male - for Personal Computer, Printer - 5 Feet (1.5 Meters), Black	695.0	4.5	10767
3	Electronics	Cameras&Photography	Accessories	💎 가성비 (실속형)	B00R1P2840	Purolite Instant Mini Single Pack 10 Sheets Instant Film for Fujifilm Instant Camera	549.0	4.5	4875
4	Electronics	HomeTheaterTV&Video	Accessories	👑 프리미엄 (하이엔드)	B08T03316Z	10k 16k 16k HDMI Cable Certified 4K@60Hz 10m Ultra High Speed HDMI 2.1 Cable 4K 120Hz 144Hz 2K 144Hz 8K 60Hz Dynamic HDR HDR10+ HDR	2399.0	4.5	3664
5	Home&Kitchen	Kitchen&HomeAppliances	SmallKitchenAppliances	👑 프리미엄 (하이엔드)	B0853DQ4TF	Instant Pot Duo 7-in-1 Electric Pressure Cooker, 6 Quart, 14.5L, 10.5L, 8.5L, 7.5L, 6.5L, 5.5L, 4.5L, 3.5L, 2.5L, 1.5L, 0.5L	20040.0	4.9	3964
6	Home&Kitchen	Kitchen&HomeAppliances	SmallKitchenAppliances	💎 가성비 (실속형)	B08BMH42N79	Multifunctional 2 in 1 Electric Egg Boiling Steamer Egg Poacher Pan Egg Boiler Electric	1599.0	4.7	2300